

tl;dr >

Planning what should be build for our internal hackathon in SIH '26, and we divided into 2 sub groups to explore 2 project ideas

Agenda:

- Domain selection for project
- Problem statement we are trying to solve
- Making a team with 6 members

Talks:

- shubham talked about their old project which is basically an on device solution for what me and swastika trying to build for our research paper.
- debayan described a very important vulnerability which no one has ever thought of mitigating for everyone, not just their own organization
- we talked about 3 essential ideas
 - shubham's idea as the fail safe backup
 - debayan's idea (active development)
 - fixing the old 2022 winner's project and take it to next level (almost no response from me)

Outcome:

- mainak is actively developing a basic impelmentation of our VPN / DNS resolver and a simulation software for that to simulate a bunch of network access logs for me to train the model tomorrow
- swastika is actively researching for me to know which deep learning model would be the best fit for this type of project.

potential team members

- ☐ roshani
- ☐ ishika
- ☒ shubham
- ☒ ritam
- ☐ arghya
- ☐ pratik

problem statement:

there might not only be threats outside of a network, the real problem might be there where no one will think of looking in the first place, where we trust the most. in the internal network of any organisation, the employees with their access can exploit those and sometimes be the reason of doom of multibillion dollar companies. how can we find the sudden change of their behaviour (i.e. totally random state of events) autonomously and take action over it.

let's say in our XYZ company there can be employess with different types of access levels, when they make any network request to our server they need to connect through our VPN so that we can log and monitor all of their activities, now among those activities if we are able to find distinguishable behavioural patterns between those activities and map them to their origin employee and their access levels, we can know for sure if any employee is doing something out of ordinary or not.

based on their behavioural patterns we can observe and monitor their changes which is way more efficient that **scouring through hours of their network logs which if you are a network engineer can understand will cause a lot of eye strain and mental stress**. we will map the change instead of daily logs of what they are doing currently, we can observe and understand the trend of the change and analyze those changes, and sometimes those anaytics can also unlock extra features like how the emplyees are doing in certain amount of time, let's say after a meeting, or after holiday, and can improve HUMAN RESOURCES strategy based on that as well

our solution:

to build an VPN / DNS resolver which will stay in the chokepoint of the whole internal network of any organization and actively monitor the changes of each employee while also activelye learning about the changes. it will also raise flags on sudden change of behaviours of any employee and if the flag is beyond a threshold of toleration then the access to the network will be cutoff for that employee until that is resolves by an admin with an higher level access.